Ni'iinlii Njik (Fishing Branch) River Chum Salmon and River Dynamic Investigations – 2023/2024

Prepared For
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ENVIRONMENTAL DYNAMICS INC.
Down to Earth Biology

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EXECUTIVE SUMMARY

Porcupine River salmon support ecosystems and peoples from the ocean to the headwaters, yet these salmon rely on relatively short stretches of headwater streams for spawning. As a result, habitat change in short stretches of streams can impact ecosystems and communities along thousands of kilometers of rivers and ocean. The Fishing Branch River (Gwitch'in: Ni'iinlii Njik) is a major spawning destination for Porcupine River chum salmon. However, in recent decades the number of returning chum salmon has been decreasing; 2020 – 2023 contained the three lowest recorded return years. In addition to the chum salmon returns being significantly lower than escapement goals, several years of study have determined that the Fishing Branch River becomes annually dewatered in spawning areas—this has resulted in up to 20% of redds dewatering and experiencing complete mortality. To better understand the proportion of redds that become dewatered over the winter and early spring, Vuntut Gwitchin First Nation (VGFN) and EDI Environmental Dynamics Inc. (EDI) completed a chum spawning aerial survey in October 2023 to enumerate redds and do maintenance on remote cameras monitoring the dewatering, and a subsequent visit in April 2024 to determine the extent to which these redds became dewatered over the winter and early spring. A visit in May 2023 is also reported on. An objective of the project is to improve our understanding of intra- and inter-annual variability of dewatering in the Fishing Branch River, specifically if this phenomenon is increasing in severity and impacting an important Indigenous fishery.

Fishing Branch River was investigated at three different time points based on project objectives. The first visit occurred on May 16, 2023 to assess the dewatered extent and its consequence to the locally impacted fish community, and complete maintenance on remote cameras. Unfortunately, due to flood conditions, these objectives couldn't be completed. An aerial survey was completed on October 28, 2023 to document spawning activity in the Fishing Branch River, from the DFO weir to ~20 km upstream, where the river is dewatered by the end of October. Remote cameras were also downloaded and redeployed for monitoring the seasonally dewatered areas. The following spring visit was on April 26, 2024 to determine the extent of the dewatered area that affected chum redd survival, and complete remote camera maintenance and downloads.

The aerial survey occurred over 1 hr 50 min and enumerated 2,402 redds, 1,432 spawning chum salmon and approximately 1,700 carcasses. The on the ground enumeration of redds is considered an overestimate due to surveyor experience and site calibration, and following QAQC, the number of redds was likely closer to 1,500. Similar to previous years, redds were clustered into three areas: CM1, CM2, and CM3. In winter and early spring of 2024, much of the CM3 area (upstream extent) became dewatered, resulting in complete redd mortality of 12% of the redds that had been enumerated in October 2023. These redd mortalities only consider redds that were completely dewatered, though it is likely that other redds are dewatered downstream of this extent but are difficult to determine at the current resolution. Future surveys intend to incorporate the use of a drone to get a finer resolution of dewatering of redds throughout the winter and early spring.

Six remote cameras were operating between April 2022 and May 2024 between three sites: FBR13/14, FBR11/22, FBR15/16. The highest upstream cameras (FBR13/14) were dewatered in every year of remote camera data, as they are farthest upstream (and closest to the October dewatered extent, which advances throughout the winter). The centrally located cameras, at site FBR11/22, FBR22 is located upstream of a

groundwater spring, and FBR11 is downstream of the same the groundwater spring. FBR11 and FBR22 were both fully dewatered in 2022, which is considered an anomalous year of dewatering as it exceeded anything in project history. In 2023 and 2024, FBR22 was fully dewatered, whereas FBR11 still had flow along the left downstream bank due to its proximity to the groundwater spring. The farthest downstream cameras, FBR15 and FBR16, were also both completely dewatered in the spring of 2022. In 2023 and 2024, the flows were greatly reduced throughout the winter and early spring, exposing some of the riverbed, though some of the river remained wetted. The 12% of redds that were dewatered between the October 2023 spawning season and May 2024 dewatering were all upstream of the spring at FBR11.

The timing of the seasonal dewatering of Fishing Branch is becoming better understood, though the processes involved are poorly understood. New questions are arising with regards to the dynamics responsible for the dewatering and the rapid return of water that occurs within 24 hr of the upstream portion of the river being completely dewatered. Future years of study will continue with fall redd assessments and subsequent spring dewatering investigations to determine the proportion of dewatered redd mortalities, to better understand the intra- and inter-annual variability of dewatering in the Fishing Branch River. In addition to continuing this work, the 2024/2025 field season intends to expand the work to better understand the processes responsible for the Fishing Branch River dewatering, with the inclusion of water sampling and hydrology monitoring.

ACKNOWLEDGEMENTS

Funding for this project was provided by the Yukon River Panel's Restoration and Enhancement Fund and helicopter funding was provided by the Polar Continental Shelf Program. All works were completed within the traditional territory of the Vuntut Gwitchin First Nation. Caleb Charlie and Jeremy Brammer (VGFN) participated for the October 2023 aerial survey. Katherine Peter and Stan Njootli Sr. (VGFN) assisted during April 2024 fieldwork and provided historical context of Fishing Branch River dynamics.

AUTHORSHIP

Team members from EDI Environmental Dynamics Inc. who contributed to preparing this report include:

Petra Szekeres, MSc, P. Biol. .... Primary Author

Ben Schonewille, R.P. Bio. .... Senior Review

Jeremy Brammer, PhD (VGFN) .... Senior Review

Phil Hesse, BA, GIS Cert. .... Mapping

Petra Szekeres (EDI), Jeremy Brammer (VGFN) and Caleb Charlie (VGFN) conducted the October 2023 aerial survey and field work. Petra Szekeres (EDI), Stan Njootli Sr. (VGFN), and Katherine Peters (VGFN) completed the April investigations. Petra Szekeres and Ben Schonewille (EDI) provided help with technical direction and logistics, while Jeremy Brammer (VGFN) provided overall project direction based on community values.

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1 INTRODUCTION

The Fishing Branch River (Gwitch'in: Ni'iinlii Njik) is a major spawning destination for Porcupine River chum salmon (*Oncorhynchus keta*). Past restoration and enhancement panel work (CRE-27-03 through CRE-27-10) indicated that in excess of 65% of chum salmon in the upper Porcupine River have been known to spawn in the upper reaches of Fishing Branch River, above the site of the Department of Fisheries and Oceans Canada (DFO) enumeration weir (Map 1). The 2011 Integrated Fisheries Management Plan developed by DFO and the Yukon River Panel included an escapement goal of 22,000 to 49,000 chum salmon at the Fishing Branch weir (JTC 2018). Since 2006, counts at the Fishing Branch weir have displayed a downward trend, often not reaching the escapement goal. This trend has been accelerated in recent years; 2020 – 2023 contained the three lowest recorded return years since 1985, when returns were first monitored (n = 4,796; n = 2,413; and n = 2,934, respectively).

Considering these declines in chum salmon productivity, the Vuntut Gwitchin First Nation (VGFN) conducted four years of chum spawning habitat assessments and monitoring between 2013 to 2019 (CRE-22-13 through CRE-163-19). A key finding of the habitat assessment work was that a portion of the Fishing Branch River becomes dewatered during the winter and early spring months, resulting in complete egg mortality for chum that spawn in this area (EDI 2014, 2018a, b). The extent of this seasonally dewatered area varies from year to year and increases in length over the course of the winter and early spring. In some years, as many as 20% of the total number of redds upstream of the enumeration weir become dewatered (EDI 2014, 2018a, b).

During the 2019/2020 season, VGFN and EDI Environmental Dynamics Inc (EDI) completed an egg incubation trial on the upper reaches of the Fishing Branch River to better understand survival and emergent timing at different locations on the river (CRE-163-19 and CRE-163-20). Broodstock were collected in the seasonally dewatered area upstream of Bear Cave Mountain, and planted at seven different locations further downstream using numerous types of planting media (EDI 2020). Results of the incubation trial had varied success across planting media and locations, and it was deemed that the approach could have merit for increasing the number of returning spawners, particularly if broodstock are collected from the area known to seasonally dewater (EDI 2020). The project was not funded by the Yukon River Panel in the 2022/2023 season, though 2022 and 2023 spring visits and remote camera data are reported on in the current report.

Since the sharp declines in returning chum salmon, the project has shifted away from instream incubation and into its current format. Particularly in light of the 2022 results, which found tens of thousands of stranded and perished freshwater fish (EDI 2021), the project objectives have shifted to better understand the hydrogeology and dynamics of this system. The focus remains on the chum spawning aerial surveys in the fall, paired with the dewatered extent in the spring, which provides an estimate of redds that become fully dewatered between the timing of spawning and maximum dewatered extent. An important note is that the purpose of the aerial survey is not to estimate escapement, as DFO operates a weir within the project area that determines escapement. Largely, the objective is to record the location and number of redds during fall spawning season, and in a spring visit, determine the proportion of these redds that become dewatered over the winter and early spring. This provides an estimate of the annual number of complete redd mortalities, and the proportion of the run that is affected across years. During 2023, collaborations were made with academic partners and government agencies to broaden the scope of this work to better understand the hydrogeology and dynamics of the system; planning was completed in 2023, with more hydro-specific elements included and reported on in upcoming 2024/2025.

The Fishing Branch River chum salmon stock is recognized as an important contributor to the Yukon River fall chum salmon run. The area is also culturally important to VGFN, who traditionally used the river as a very reliable fall fishing location in preparation for winter. It was this knowledge of the importance of the Fishing Branch River for spawning chum that motivated the First Nation to argue forcefully for the inclusion of protected areas for the Fishing Branch River in their land claim, settled with the government of Canada in 1993. As a result, the Fishing Branch watershed is entirely protected through four land designations: Vuntut Gwitchin First Nation Settlement Lands, an Ecological Reserve, a Wilderness Reserve, and a Habitat Protection Area. Overall, this project will provide information that is important to the effective, informed management of the Fishing Branch River chum salmon stock and particularly its spawning habitat in a changing environment. Effective management of this chum stock should ensure it remains an important contributor to the economic and cultural wellbeing of the VGFN and the health and wellbeing of ecosystems and peoples down the entire Yukon River watershed and on the Bering Sea.

The priority status of the Fishing Branch River stock and the activities proposed by this project are directly referred to under Section 26 of the Yukon Salmon Agreement (2001). This section of the agreement states:

"26. With respect to the Fishing Branch River chum salmon, the Parties agree that when spawning escapements fall below target levels for this stock as specified in Appendix 1 to Attachment B, the Yukon River Panel shall consider the need to develop a rebuilding plan based on information and analysis from the JTC. If the Yukon River Panel decides that such a plan is needed, it shall request the JTC to prepare a range of rebuilding plan options, including allowing this stock to rebuild as a result of the rebuilding program for the Yukon River Mainstem fall chum salmon stock. The Panel shall determine which plan to recommend to the respective management entities."

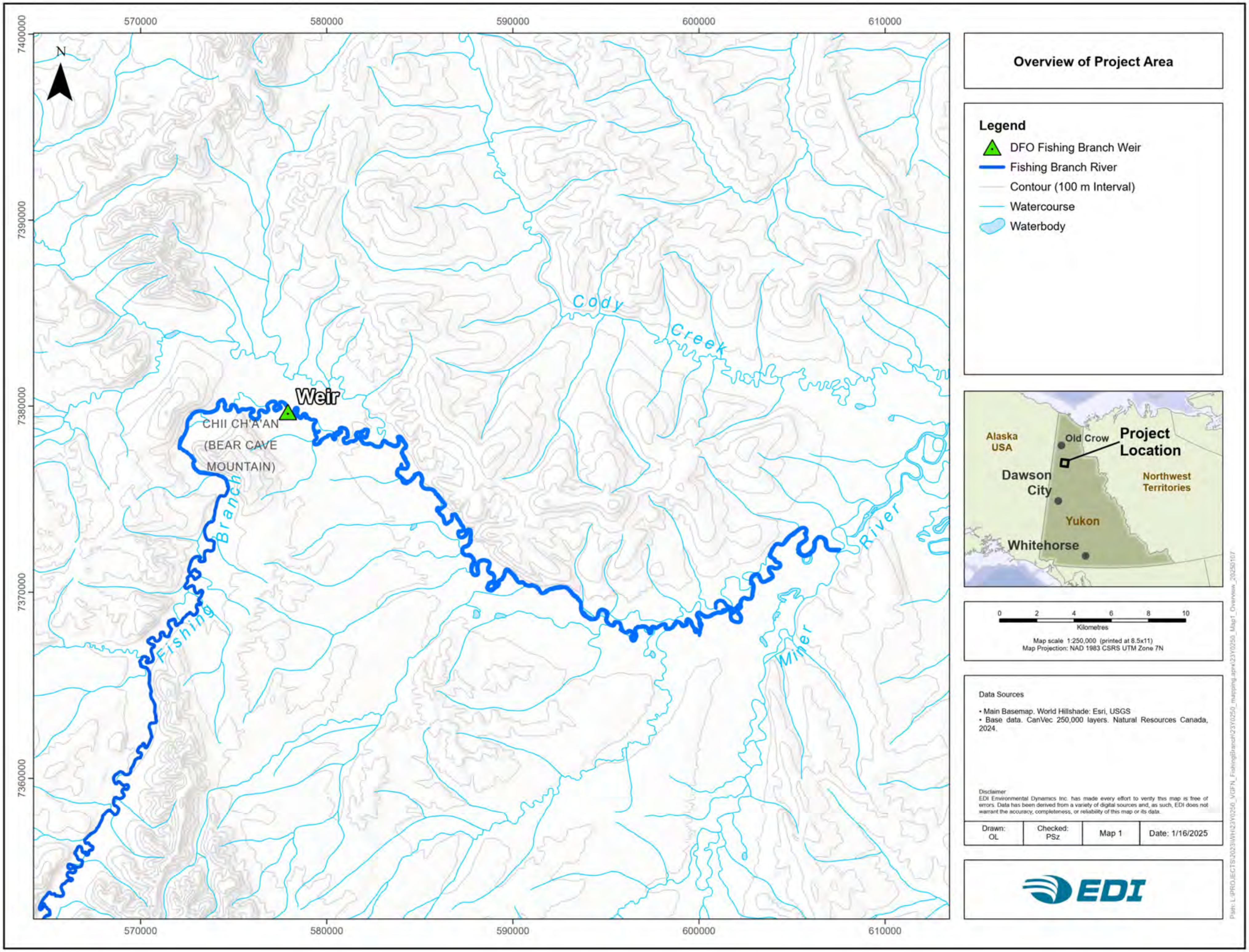
This project will improve our understanding of intra and inter-annual variability of dewatering in the Fishing Branch River, specifically if this phenomenon is increasing in severity and impacting an important Indigenous fishery.

The 2023/2024 project activities included three site visits. In response to record low chum spawner returns, VGFN has redirected efforts to further understand the extent and timing of the seasonally dewatered area and its impacts on chum spawner success. The first visit was May 16, 2023 to determine the dewatered extent, as the previous April/May 2022 visit provided unanticipated results with regards to fish stranding and a mass mortality event, prompting further attention to this time of the year. The following visit occurred on October 28, 2023 and involved an aerial survey to enumerate redds, spawning chum, and carcasses, and complete maintenance of remote cameras that have been installed in the seasonally dewatered area to document conditions and the extent of the dewatered area throughout the year. A trip on April 26, 2024 included a brief investigation of the dewatered extent and maintenance of remote cameras.

1.1 OBJECTIVES

The Fishing Branch River objectives were to:

- Conduct an aerial survey to enumerate spawning chum salmon, redds and carcasses on the Fishing Branch River from just downstream of the DFO weir and ending approximately 20 river km upstream at a point where the river is dewatered;
- Determine dewatered extent in early spring to estimate number of dewatered redds over the winter;
- In spring investigate dewatered areas and isolated pools for fish stranding;
- Download and redeploy remote cameras to further investigate the timing and extent of the seasonal dewatered areas in the Fishing Branch River.



Overview of Project Area
Legend
DFO Fishing Branch Weir
Fishing Branch River
Contour (100 m Interval)
Watercourse
Waterbody
570000
580000
590000
600000
610000
7400000
7390000
7380000
7370000
7360000
N
Weir
CHII CH'A'AN
(BEAR CAVE
MOUNTAIN)
Cody
Creek
Fishing
Branch
River
Miner
Alaska
USA
Old Crow
Project
Location
Dawson
City
Northwest
Territories
Yukon
Whitehorse
0
2
4
6
8
10
Kilometres
Map scale 1:250,000 (printed at 8.5x11)
Map Projection: NAD 1983 CSRS UTM Zone 7N
Data Sources
• Main Basemap. World Hillshade: Esri, USGS
• Base data. CanVec 250,000 layers. Natural Resources Canada, 2024.
Disclaimer
EDI Environmental Dynamics Inc. has made every effort to verify this map is free of errors. Data has been derived from a variety of digital sources and, as such, EDI does not warrant the accuracy, completeness, or reliability of this map or its data.
Drawn: OL
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Map 1
Date: 1/16/2025

2 METHODS

Fishing Branch River was investigated at three different time points based on project objectives. The first visit occurred on May 16, 2023 to assess the dewatered extent and its consequence to the locally impacted fish community, and complete maintenance on remote cameras. The following visit was on October 28, 2023 to complete an aerial chum spawning survey, and download and complete maintenance on remote cameras for monitoring the seasonally dewatered area. The following spring visit was on April 26, 2024 to determine the extent of the dewatered area, complete on the ground investigations, and remote camera maintenance and downloads. All distances are provided as river kilometers (i.e., following the meandering of the river), not 'as the crow flies' (i.e., straight-line land measurements).

2.1 CHUM SALMON REDD AND SPAWNER AERIAL SURVEY COUNTS

On October 28, 2023, the field crew conducted a low altitude helicopter based aerial survey to enumerate spawning chum salmon and redd locations within the Fishing Branch River study area. The focus of the survey was to determine the spatial distribution and abundance of chum salmon redds and spawners, including in tributary and side channels. The survey was intentionally conducted after the peak chum salmon spawning had occurred to observe the highest possible number of constructed redds; enumeration of spawning chum salmon is of lower priority to this study. A more accurate measure of the 2023 spawner escapement is provided by the counts collected at the DFO Fishing Branch River weir (Alaska Department of Fish and Game 2024)

A single pass of the river was completed beginning at the DFO weir and ending approximately 20 river km upstream at a point where the river is dewatered in October. During the survey, observers were in the front and back seats on the passenger side of the helicopter looking over the river, with the best possible vantage point at approximately 30 – 45 m above ground level and flying 10 to 15 knots, depending on visibility. The survey was flown in an upstream direction to reduce glare. All observed redds, spawning chum salmon, and carcasses were enumerated, and their locations recoded using a GPS. An iPad and the mapping application Avenza Pro were used to direct the spawning survey extent and collect data. Where redds and salmon were clustered, a single GPS waypoint was recorded and estimates of the number of redds and salmon were made. The survey covered 20 river km and took 1 hr 55 min.

An important note on the 2023 results. While the surveying was thorough and had good visibility, and had no constraints related to daylight, weather, or fuel, due to the calibration required of an observer who is new to this river system and species, redd numbers may be erroneously high. Thus, results are provided as they were observed on the ground, and as calibrated/QAQC'd by someone very familiar with this system. This was completed by the QAQC person estimating redds from a series of different photos, and comparing to what was recorded in the field. In the field estimates were approximately 35% higher than QAQC amounts; both are provided in the results. However, what is often most interesting with these results is not necessarily the specific numbers, but the trends/proportions of chum focused in certain high density areas, and the proportion of redds dewatered over the winter.

2.2 REMOTE CAMERA MONITORING AND SPRING DEWATERED EXTENT INVESTIGATIONS

During the 2021/2022 project, a total of six remote cameras were deployed between Bear Cave Mountain and where the river has no flow in the fall, approximately 4.5 river km upstream of Bear Cave Mountain. The intent is to complete maintenance and download these cameras during each visit, though this is not always possible due to time/fuel/environmental constraints. Remote cameras are programmed to take six photos a day, in half hour intervals beginning at 12:00 and ending at 14:30.

Fishing Branch River was revisited on May 16, 2023, and April 26, 2024 to investigate the seasonally dewatered extent observed during past years of study (EDI 2018b, 2021). These visits are completed to pair with the preceding October spawning survey results to assess the number of October chum redds that become dewatered over the late winter and early spring.

The May 16, 2023 visit missed the dewatered extent window and freshet had occurred within 24 hr of arrival, with flood conditions (Photo 1). Due to the high water levels, no landing sites were available to access the remote camera sites, thus no cameras were downloaded during this visit. No dewatered extent was identified on the ground during this visit, and instead satellite imagery was used to estimate dewatered extent and timing. Copernicus Browser (European Space Agency 2025) was used to determine when freshet occurred, and how far the dewatered extent was leading up to freshet. This information was paired with results from the remote cameras.

The October 28, 2023 chum salmon aerial surveying allowed for a brief window to download and redeploy cameras FBR13, FBR14, FBR15, and FBR16. These cameras had their storage backed up, and batteries exchanged for new ones.

The April 26, 2024 site visit was very brief due to pilot duty day restrictions and the model of helicopter used, resulting in limitations due to fuel and daylight constraints. Only a single helicopter landing and take off was possible, thus FBR11, FBR22, FBR15, and FBR16 were accessed on foot and there was only time to exchange the batteries and memory cards. Time did not allow for maintenance, checking settings or positioning. No detailed investigations were possible regarding fish stranding, as the limited time on the ground was spent completing camera downloads and redeployment. Similarly, satellite imagery was used to support and confirm what field crews observed.

Photo 1. Fishing Branch River in flood on May 16, 2023.

3 RESULTS AND DISCUSSION

October 2023 aerial survey results and discussion are presented below, in addition to 2022 – 2024 remote camera data from six cameras recording photos on Fishing Branch River dynamics throughout the year.

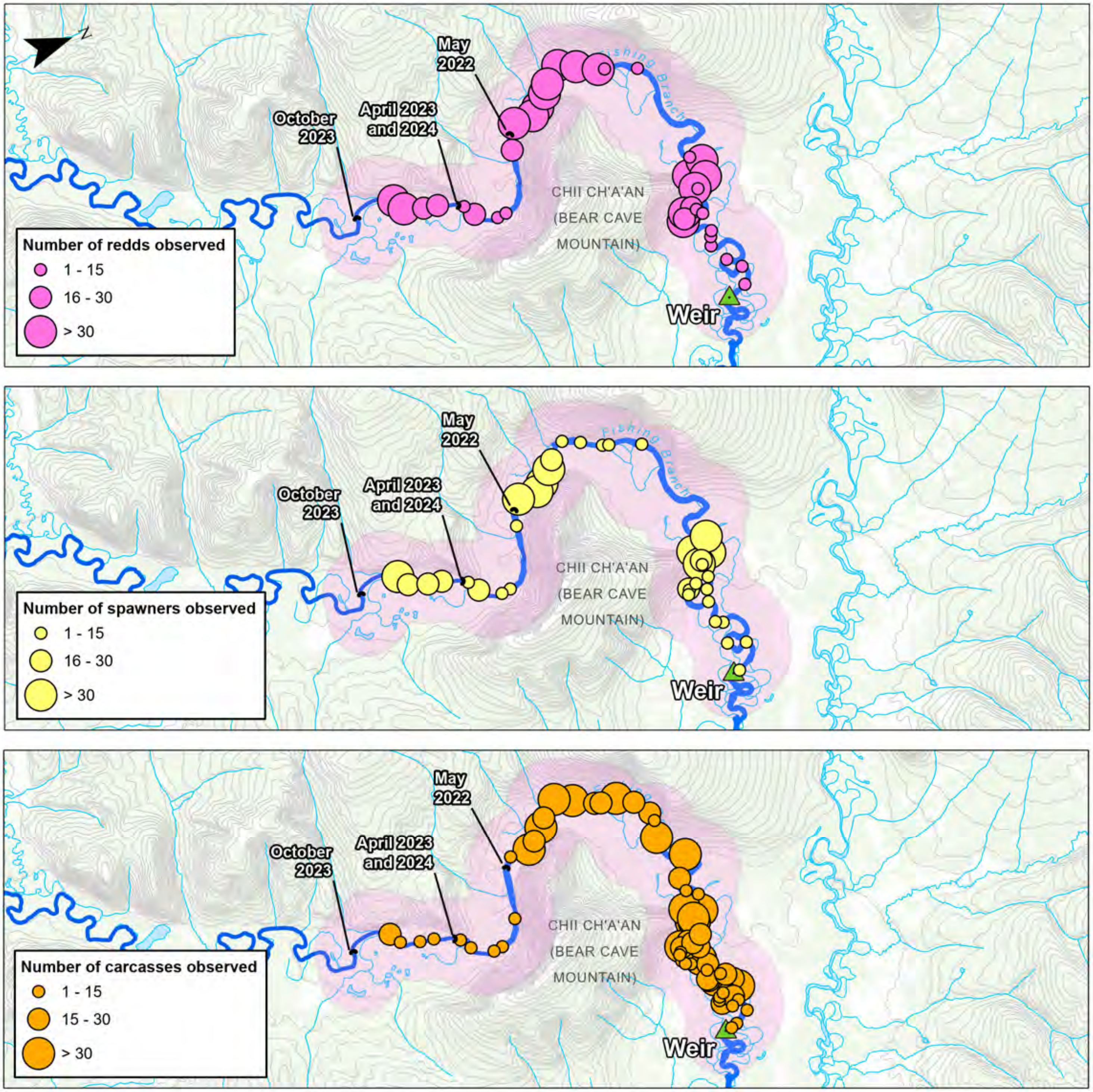
3.1 OCTOBER AERIAL CHUM SPAWNING SURVEY

The purpose of the chum salmon spawning aerial survey is to get an estimate of redds near the end of the spawning season, such that in the subsequent spring visit there can be an estimate of the number and proportion of redds that became dewatered over the course of the late winter and early spring. The aerial survey is not meant (or timed) to get an escapement estimate, as the DFO weir is designed to gather this information, far more accurately than a late season aerial survey. The DFO weir counted 11,528 returning chum salmon in 2023 (Alaska Department of Fish and Game 2024).

In contrast to logistical challenges faced in other years (e.g., EDI 2021), the October 28, 2023 aerial survey was completed on the day it was planned with no weather delays and no constraints related to fuel, as fuel had been cached at the DFO camp prior to surveying. Weather conditions during the October 28, 2023 aerial survey were overcast, with temperatures ranging from -10°C to -15°C. Visibility was considered good, with helicopter positioning/flying adjustments used to offset the glare from overcast conditions. The total time to complete the aerial redd count was approximately 1 hr 50 min.

A total of 2,402 redds were identified during the survey (Photo 2, Photo 3; Map 2, Map 3; Appendix Table 1). As previously mentioned, the on the ground estimates were QAQC'd and are considered a high estimate. Adjusted by an experienced surveyor, field estimates were approximately 35% higher than QAQC'd amounts. Therefore, calibrated against this overestimation, redd counts were likely closer to 1,560 in the surveyed extent. For simplicity and consistency, data will primarily be discussed as it was observed on the ground, with the understanding that it is considered an overestimation.

Spawning was observed throughout the study area, including within the upper portion of the river that has been documented to become seasonally dewatered in previous years (Map 2, Map 3; EDI 2014, EDI 2016, EDI 2018a, b, 2021). Similar to the redd distribution documented during the past aerial chum surveys conducted by VGFN and EDI (EDI 2014, EDI 2016, EDI 2018a, b, 2021), redds were predominately clustered in three main chum salmon spawning areas (Map 2, Map 3).



Legend

- DFO Fishing Branch Weir
- Fishing Branch River
- Dewatered Extents
- Contour (20 m Interval)
- Watercourse
- Waterbody
- Survey Area

Overview of chum spawning aerial survey results

Data Sources

- Main Basemap. World Hillshade: Esri, USGS
- Base data. CanVec 250,000 layers. Natural Resources Canada, 2024.
- Salmon observations. EDI Environmental Dynamics Inc, 2024.

0 2 4 6
Kilometres

Map Scale: 1:120,000 (printed on 8.5 x 11)
Map Projection: NAD 1983 CSRS UTM Zone 7N
Map Rotation: -70°

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EDI

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Fishing Branch River Chum Salmon Redd Count Data - October 2023

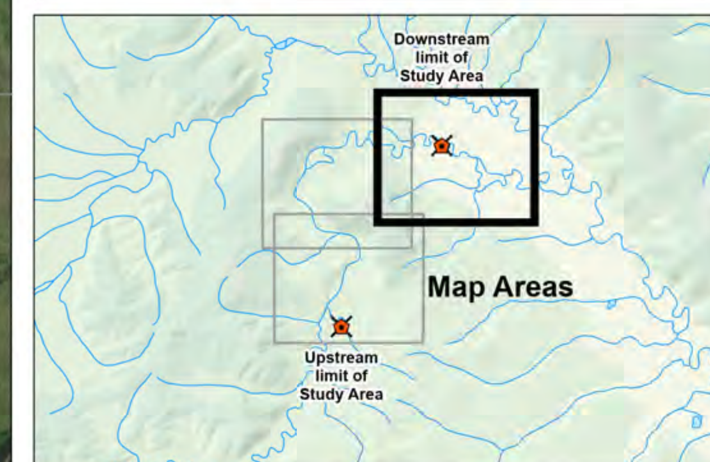
Map 1 of 3

Legend

- Limit of Study Area
- DFO Fishing Branch Weir
- Dewatered Extents

Quantity of Chum Salmon Redds

- 1-25
- 26-75
- 76-150
- 151-250
- 251-370



0 250 500 750 1,000

Metres

Map Scale = 1:20,000 (printed on 11 x 17)
Map Projection: NAD 1983 CSRS UTM Zone 7N

Data Sources
- Main Basemap. World Imagery: Maxar
- Main data. CanVec 1:250,000 layers. Natural Resources Canada, 2024.
- Salmon observations. EDI Environmental Dynamics Inc, 2024.

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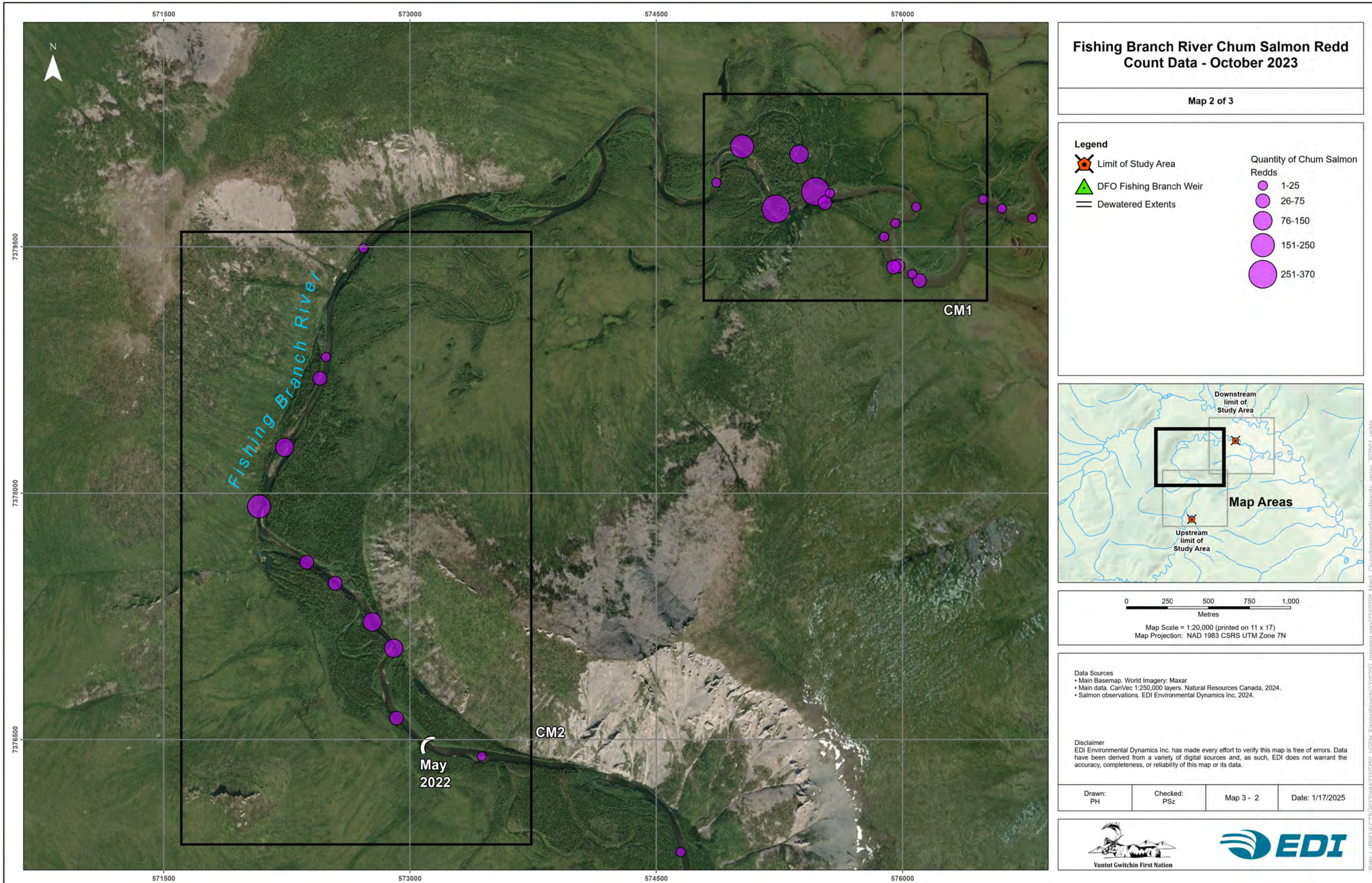
Vuntut Gwitchin First Nation

EDI



Fishing Branch River Chum Salmon Redd Count Data - October 2023
Map 2 of 3
Legend
Limit of Study Area
DFO Fishing Branch Weir
Dewatered Extents
Quantity of Chum Salmon Redds
1-25
26-75
76-150
151-250
251-370
Downstream limit of Study Area
Map Areas
Upstream limit of Study Area
0
250
500
750
1,000
Metres
Map Scale = 1:20,000 (printed on 11 x 17)
Map Projection: NAD 1983 CSRS UTM Zone 7N
Data Sources
• Main Basemap. World Imagery: Maxar
• Main data. CanVec 1:250,000 layers. Natural Resources Canada, 2024.
• Salmon observations. EDI Environmental Dynamics Inc, 2024.
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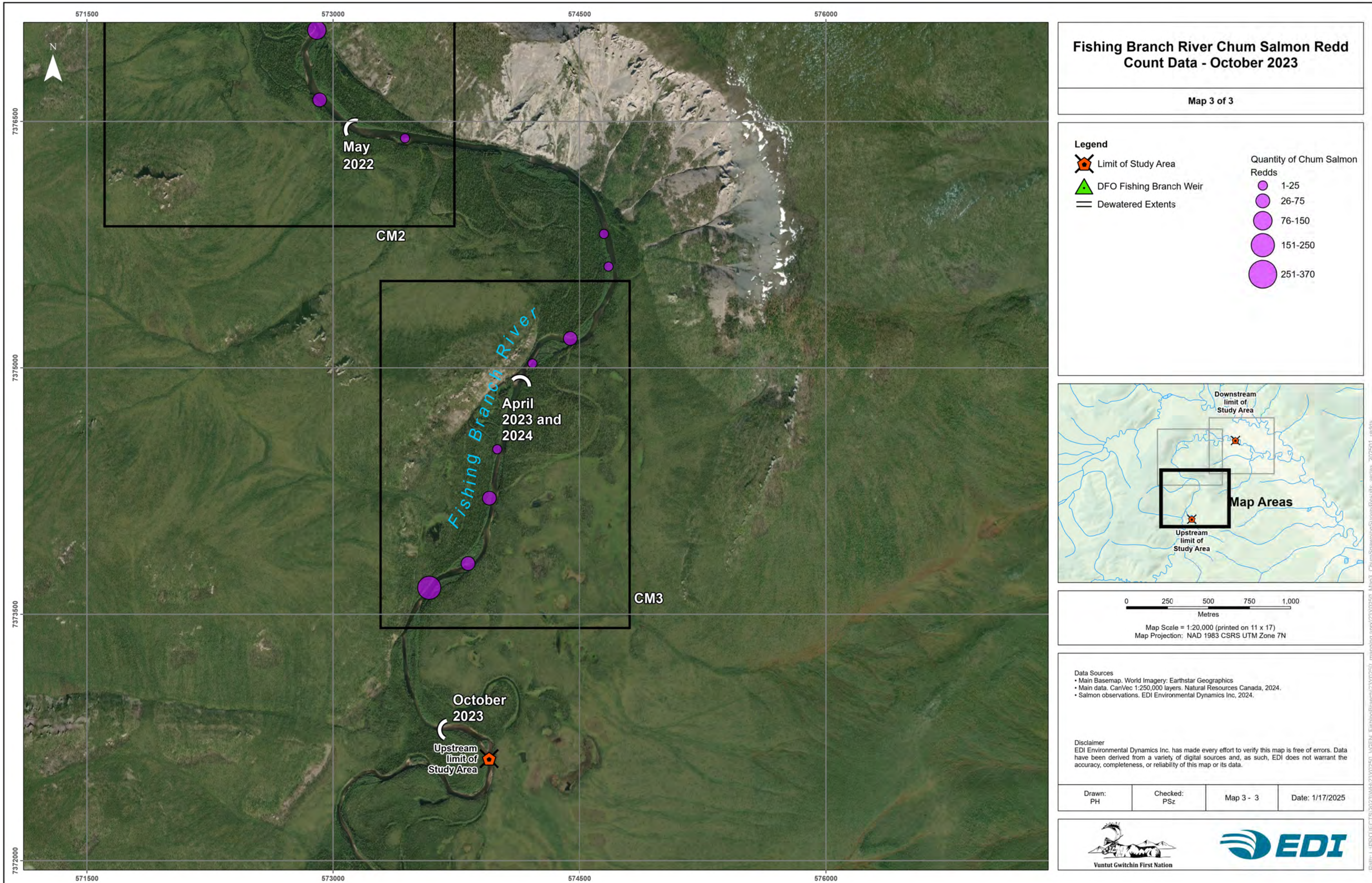
Map 3 - 2
Date: 1/17/2025
Vuntut Gwitchin First Nation
N
571500
573000
574500
576000
7379500
7378000
7376500
Fishing Branch River
CM1
CM2
May 2022


Fishing Branch River Chum Salmon Redd Count Data - October 2023
Map 3 of 3
Legend
Limit of Study Area
DFO Fishing Branch Weir
Dewatered Extents
Quantity of Chum Salmon Redds
1-25
26-75
76-150
151-250
251-370
Downstream limit of Study Area
Map Areas
Upstream limit of Study Area
0
250
500
750
1,000
Metres
Map Scale = 1:20,000 (printed on 11 x 17)
Map Projection: NAD 1983 CSRS UTM Zone 7N
Data Sources
• Main Basemap. World Imagery: Earthstar Geographics
• Main data. CanVec 1:250,000 layers. Natural Resources Canada, 2024.
• Salmon observations. EDI Environmental Dynamics Inc, 2024.
Disclaimer
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Drawn: PH
Checked: PSz
Map 3 - 3
Date: 1/17/2025
Vuntut Gwitchin First Nation
EDI
N
571500
573000
574500
576000
7376500
7375000
7373500
7372000
May 2022
CM2
Fishing Branch River
April 2023 and 2024
CM3
October 2023
Upstream limit of Study Area

Spawning in areas CM1, CM2, and CM3 accounted for 98.4% of the total observed chum salmon redds documented in October 2023, similar to previous years (Table 1; Map 3). These three clusters are visibly apparent (Map 2, Map 3) and coincide with groundwater inputs in the river (Utting et al. 2012, EDI 2016, 2018a, b). As fuel and other field constraints have become more well-known throughout the past years of study on Fishing Branch River, the length of the aerial surveys have not been consistent (e.g., 2017 survey began further downstream than 2023), thus the number of redds in "other areas" may be representative of different lengths of surveyed areas (Table 1).

Table 1. Distribution of redds and the relative proportion of total spawning during the 2013, 2015 – 2017, and 2023 aerial redd surveys.

Spawning Area Name	Total Number of Redds (*n* =) and Relative Proportion of Total Spawning (% of total redds)											
	2013		2015		2016		2017		2021[A]		2023[B]	
	n =	%	*n* =	%	*n* =	%	*n* =	%	*n* =	%	*n* =	%
CM1	406	35.6	700	52.0	268	31.5	936	29.8	-	-	1,262	52.5
CM2	467	41.0	589	43.7	331	38.9	1,233	39.3	-	-	779	32.4
CM3	157	13.8	43	3.2	240	28.2	719	22.9	-	-	323	13.4
Other Areas	110	9.6	14	1.0	12	1.4	251	8.0	-	-	38	1.6
Total	1,140	-	1,346	-	851	-	3,139	-	-	-	2,402	-

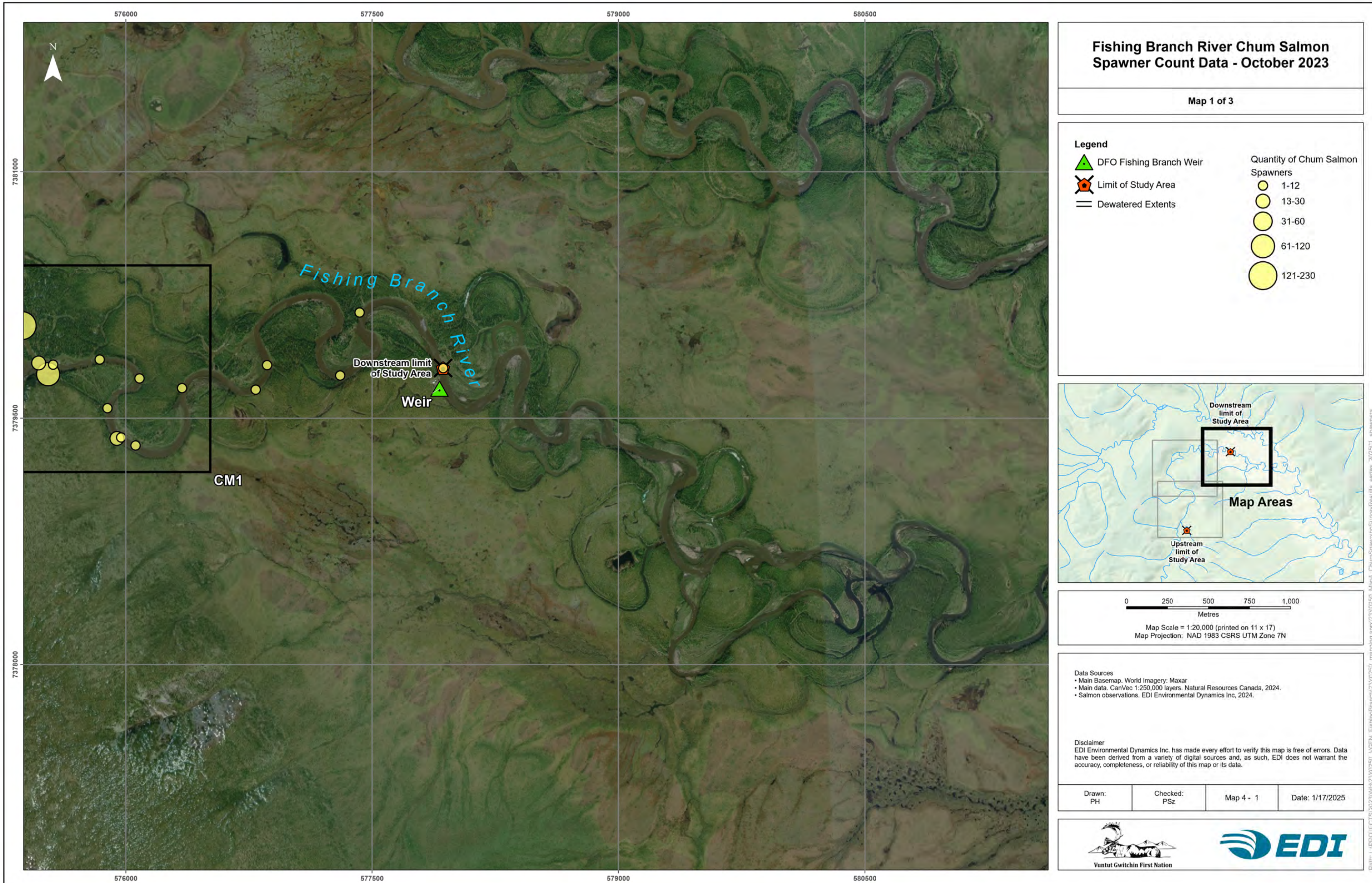
[A] Due to fuel and daylight constraints, the survey wasn't completed in a way that allows for year-to-year comparisons.
[B] These numbers are as they were recorded on the ground and have not been adjusted through office QAQC; therefore, these numbers are an overestimation. Accounting for QAQC and the overestimation, from top to bottom the adjusted 2023 *n* = are: 820, 506, 210, 25, and 1,561. The percentages remain the same.

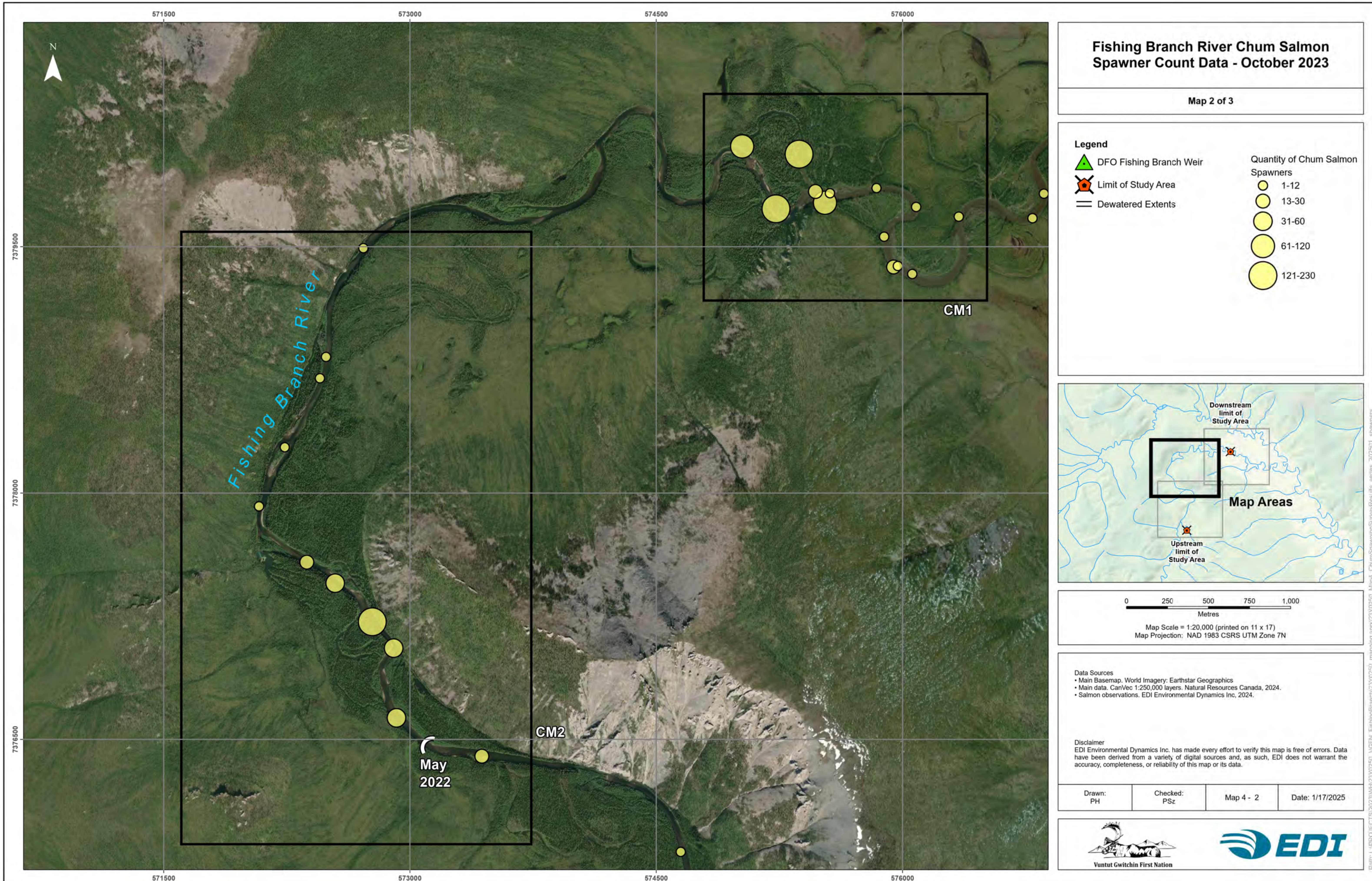
Redds are often clustered densely, with past works on Fishing Branch River illustrating that chum will often dig up existing redds and deposit their own eggs. Due to the high density of redds, it is often difficult to distinguish where one redd ends and another begins (Photo 2, Photo 3), and can result in erroneously high estimates without prior surveying experience in this system. The surveyor has now been calibrated against a biologist with extensive experience in this watershed, and there are no concerns for consistency and accuracy in future surveys. Fishing Branch chum redds are assumed to follow results from chum redds in the Columbia River watershed, which are approximately 2.25 m^2 (~1.5 m × 1.5 m in dimension; Burner 1951).

Spawner locations were also tallied and location recorded (Map 4; Appendix Table 1). The October 2023 aerial survey estimated 1,432 spawning salmon and approximately 1,700 carcasses (Map 2, Map 4; Appendix Table 1). While carcasses were observed fairly consistently throughout the aerial survey extent (likely due to current washing carcasses downstream and redistributing them), spawners and redds followed a similar clustering pattern (Map 2, Map 4).

Photo 2. Chum salmon redds adjacent to a side channel in area CM1 observed on October 28, 2023.

Photo 3. Chum salmon redds and carcasses mid-channel in area CM1 observed on October 28, 2023.


Fishing Branch River Chum Salmon Spawner Count Data - October 2023
Map 1 of 3
Legend
DFO Fishing Branch Weir
Limit of Study Area
Dewatered Extents
Quantity of Chum Salmon Spawners
1-12
13-30
31-60
61-120
121-230
Fishing Branch River
Downstream limit of Study Area
Weir
CM1
N
576000
577500
579000
580500
7381000
7379500
7378000
Downstream limit of Study Area
Upstream limit of Study Area
Map Areas
0
250
500
750
1,000
Metres
Map Scale = 1:20,000 (printed on 11 x 17)
Map Projection: NAD 1983 CSRS UTM Zone 7N
Data Sources
• Main Basemap. World Imagery: Maxar
• Main data. CanVec 1:250,000 layers. Natural Resources Canada, 2024.
• Salmon observations. EDI Environmental Dynamics Inc, 2024.
Disclaimer
EDI Environmental Dynamics Inc. has made every effort to verify this map is free of errors. Data have been derived from a variety of digital sources and, as such, EDI does not warrant the accuracy, completeness, or reliability of this map or its data.
Drawn: PH
Checked: PSz
Map 4 - 1
Date: 1/17/2025
Vuntut Gwitchin First Nation
EDI
Path: L:\PROJECTS\2023\WH\23Y0250_VGFN_FishingBranch\23Y0250_mapping.aprx\23Y0250_Map4_ChumSpawningSurveyResults_series_202501_spawners



Fishing Branch River Chum Salmon Spawner Count Data - October 2023
Map 2 of 3
Legend
DFO Fishing Branch Weir
Limit of Study Area
Dewatered Extents
Quantity of Chum Salmon Spawners
1-12
13-30
31-60
61-120
121-230
N
571500
573000
574500
576000
7379500
7378000
7376500
Fishing Branch River
CM1
CM2
May 2022
Downstream limit of Study Area
Upstream limit of Study Area
Map Areas
0
250
500
750
1,000
Metres
Map Scale = 1:20,000 (printed on 11 x 17)
Map Projection: NAD 1983 CSRS UTM Zone 7N
Data Sources
• Main Basemap. World Imagery: Earthstar Geographics
• Main data. CanVec 1:250,000 layers. Natural Resources Canada, 2024.
• Salmon observations. EDI Environmental Dynamics Inc, 2024.
Disclaimer
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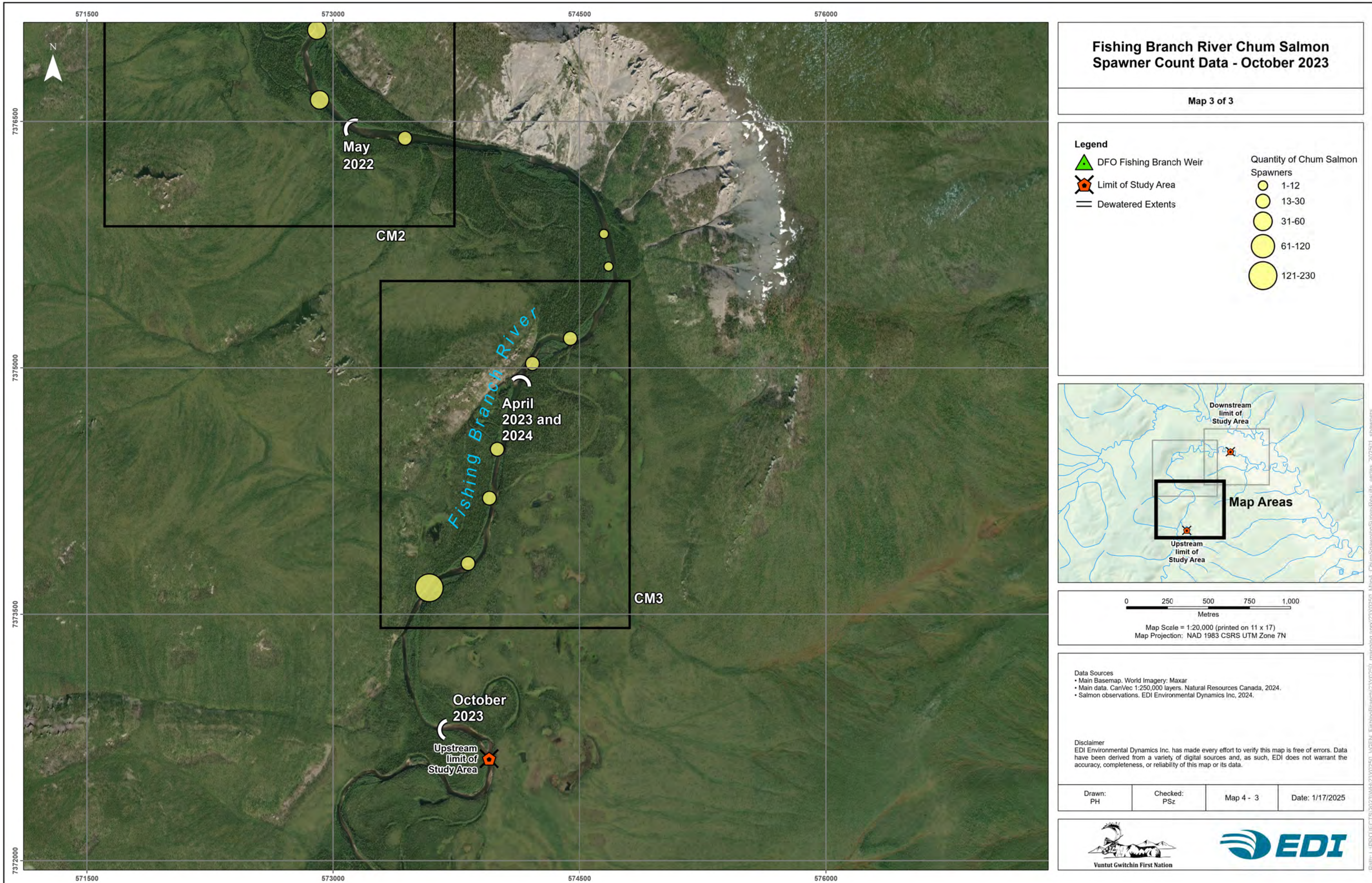
Drawn: PH
Checked: PSz
Map 4 - 2
Date: 1/17/2025
Vuntut Gwitchin First Nation
EDI
Path: L:\PROJECTS\2023\WH23Y0250_VGFN_FishingBranch\23Y0250_mapping.aprx\23Y0250_Map4_ChumSpawningSurveyResults_series_202501_spawners



Fishing Branch River Chum Salmon Spawner Count Data - October 2023
Map 3 of 3
Legend
DFO Fishing Branch Weir
Limit of Study Area
Dewatered Extents
Quantity of Chum Salmon Spawners
1-12
13-30
31-60
61-120
121-230
Downstream limit of Study Area
Map Areas
Upstream limit of Study Area
0 250 500 750 1,000
Metres
Map Scale = 1:20,000 (printed on 11 x 17)
Map Projection: NAD 1983 CSRS UTM Zone 7N
Data Sources
• Main Basemap. World Imagery: Maxar
• Main data. CanVec 1:250,000 layers. Natural Resources Canada, 2024.
• Salmon observations. EDI Environmental Dynamics Inc, 2024.
Disclaimer
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Drawn: PH
Checked: PSz
Map 4 - 3
Date: 1/17/2025
Vuntut Gwitchin First Nation
EDI
N
571500
573000
574500
576000
7376500
7375000
7373500
7372000
May 2022
CM2
Fishing Branch River
April 2023 and 2024
CM3
October 2023
Upstream limit of Study Area

Redds in CM1 and CM2 (Map 3) have historically been outside of the dewatered zone (with the exception of 2022; EDI 2021), with groundwater inputs from the aquifer adjacent to Bear Cave Mountain ensuring flow throughout the year (Utting et al. 2012, EDI 2016). However, essentially all redds in CM3 are at risk of dewatering, especially if the river dewaters to the extreme extent of May 2022 (Map 3).

Even though CM3 has the fewest redds proportionally upstream of the DFO weir (Table 1), it still makes up an average of 16.3 ± 3.8% SE of the proportion of redds found in the surveyed area since 2013 (5 years of data; Table 1). The October 2023 spawning results indicate that 12% of the redds that were observed in the fall were completely dewatered by May 2024, all within CM3. Project experience and past studies on Fishing Branch River have determined that there aren't sufficient subsurface flows to sustain redds and egg development, and that all redds in the fully dewatered area have no survival (EDI 2018b).

The upstream section of the project area becomes dewatered in most years, covering a large area of CM3 and several clusters of redds in most years. All redds on Map 3 that are upstream of the "April 2023 and 2024" label became fully dewatered from October 2023 to May 2024. Based on the on the ground estimates, this equates to 288 redds that were completely dewatered and had complete redd mortality (187 redds based on the QAQC'd estimates). It is estimated that in North America each female chum spawns 2000 – 4000 eggs per redd (Groot and Margolis 1991). Therefore, using the QAQC value of 187 redds in this dewatered area, it can be estimated that somewhere between 350,000 to 750,000 eggs were frozen and died over the course of the 2023/2024 winter and early spring.

During the 2023 survey there was an area of particularly high density redds and spawners just downstream of where the river was dewatered in October 2023, amounting to 202 redds (QAQC amount 131 redds; Photo 4). This area was fully dewatered by March 2024, based on remote camera imagery discussed in the below section (Section 3.2); this area has had redds dewatered in all years that it has been studied (EDI 2014, EDI 2016, EDI 2018a, b, 2021). In October 2023, the river was dewatered 4.5 km upstream of Bear Cave Mountain (Map 4, Photo 5). From this location, it continues to dewater downstream throughout the winter into the early spring (detailed in Section 3.2). Typically, and as was the case in May 2024, it becomes fully dewatered downstream to approximately camera FBR22 (Map 4, Map 5).

Though areas that fully dewater make it apparent that redds experienced complete mortality, areas that remain wetted but with significantly reduced flows may also have redd mortality that is more difficult to quantify. This is discussed in further detail in the below section (Section 3.2) and has prompted some planning on how to better quantify redd loss in areas that have reduced flows, but aren't completely dewatered from bank to bank. Future project years will likely consider the use of a drone to better assess the location of redds during fall spawning (October) and fly the same extent again in the early spring (April/May) to create an overlay of redds that exist in areas that become dewatered over the winter.

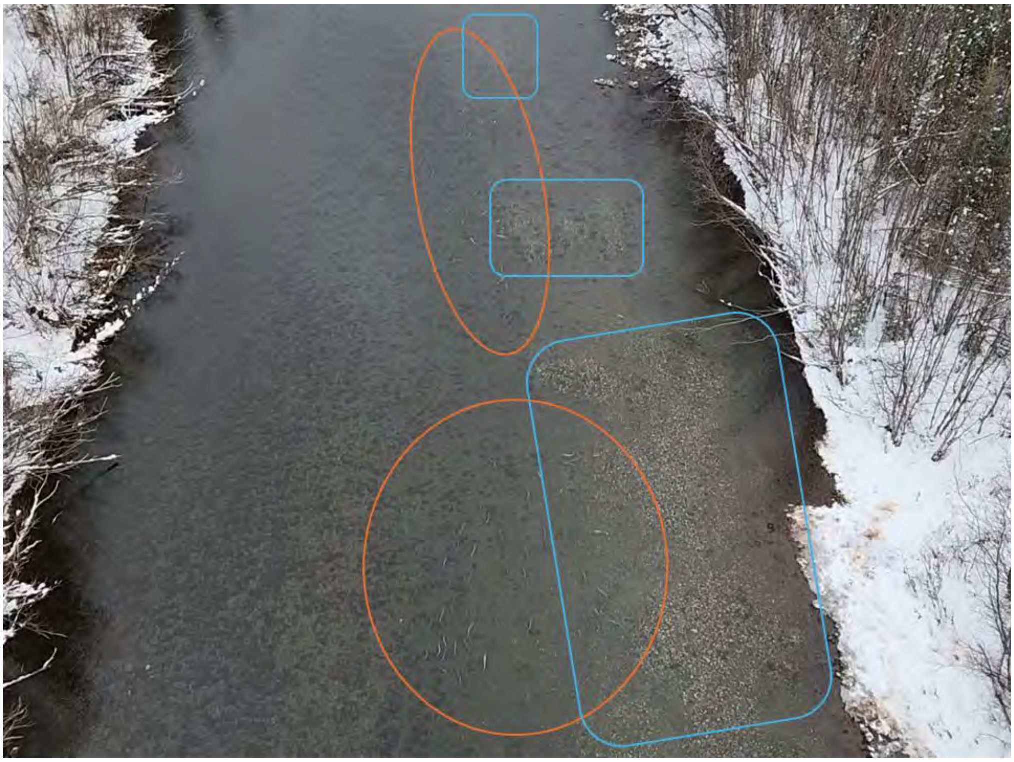


Photo 4. High numbers of spawning chum (orange circles) and redds (blue rectangles) on October 28, 2023, with recent feeding on carcasses on the river margins (e.g., directly to the right of the largest blue rectangle). All the redds pictured dewatered in winter and early spring 2024.

Photo 5. Downstream facing photo of the location that the river was dewatered on October 28, 2023.

3.2 REMOTE CAMERA MONITORING AND SPRING DEWATERED EXTENT INVESTIGATIONS

Six remote cameras were deployed during the 2021/2022 field season. Two cameras, FBR11 and FBR22, installed in October 2021, were redeployed in April 2022, while four additional cameras (FBR13 to FBR16) were newly deployed during the same April visit. Cameras FBR13 and FBR14 were positioned upstream of FBR11 and FBR22, whereas cameras FBR15 and FBR16 were set downstream (Map 5). Data from all cameras was retrieved and downloaded on October 28, 2023 and again October 31, 2024, and is detailed in the following report. The reported data spans from April 29, 2022 to October 31, 2024. Cameras were deployed on April 29, 2022 and recorded six photos per day until they were downloaded during subsequent visits, as time and fuel allowed. Photos were taken in half hour intervals beginning at 12:00 and ending at 14:30.

Due to transfer errors or wildlife encounters, not all cameras had a complete time series between April 2022 and October 2024. At each site location/camera pairing, the camera with the most complete time series is reported on in more detail. During the 2024/2025 field season, lock boxes will be installed onto each camera, and more time devoted to maintenance on all cameras. There have been time and fuel constraints in past consecutive field visits, resulting in cameras not being downloaded or maintained as frequently as is ideal.

The 2024 spring dewatered extents are used in combination with the 2023 fall redd counts to determine the number of redds that become dewatered and experience complete redd mortality. The remote camera data presented for 2022 and 2023 cannot be discussed accurately with their preceding spawning surveys; this is due to the October 2021 survey being flown fast due to fuel constraints, resulting in less accurate counts, and the October 2022 survey not occurring due to funding.

The 2022/2023 project was not funded by the Yukon River Panel, therefore the remote camera data from 2022/2023 has not been reported on. In subsequent visits these cameras were downloaded, and their results are shared in this report, in addition to the 2023/2024 results.

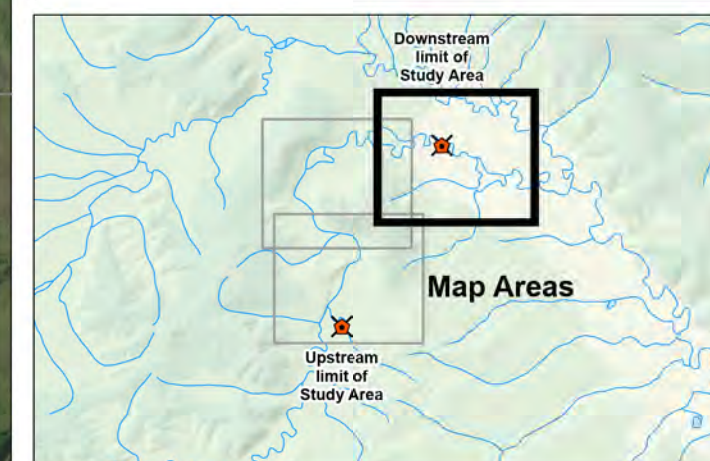
FBR13 was operational most consistently of all the cameras, thus there is additional data presented on this most upstream camera (Appendix C).

Fishing Branch River camera monitoring sites and spring dewatering locations

Map 1 of 3

Legend

- Limit of Study Area
- DFO Fishing Branch Weir
- Dewatered Extents
- Camera Monitoring Location



0 250 500 750 1,000
Metres

Map Scale = 1:20,000 (printed on 11 x 17)
Map Projection: NAD 1983 CSRS UTM Zone 7N

Data Sources
• Main Basemap. World Imagery: Maxar
• Main data. CanVec 1:250,000 layers. Natural Resources Canada, 2024.
• Salmon observations. EDI Environmental Dynamics Inc, 2024.

Disclaimer
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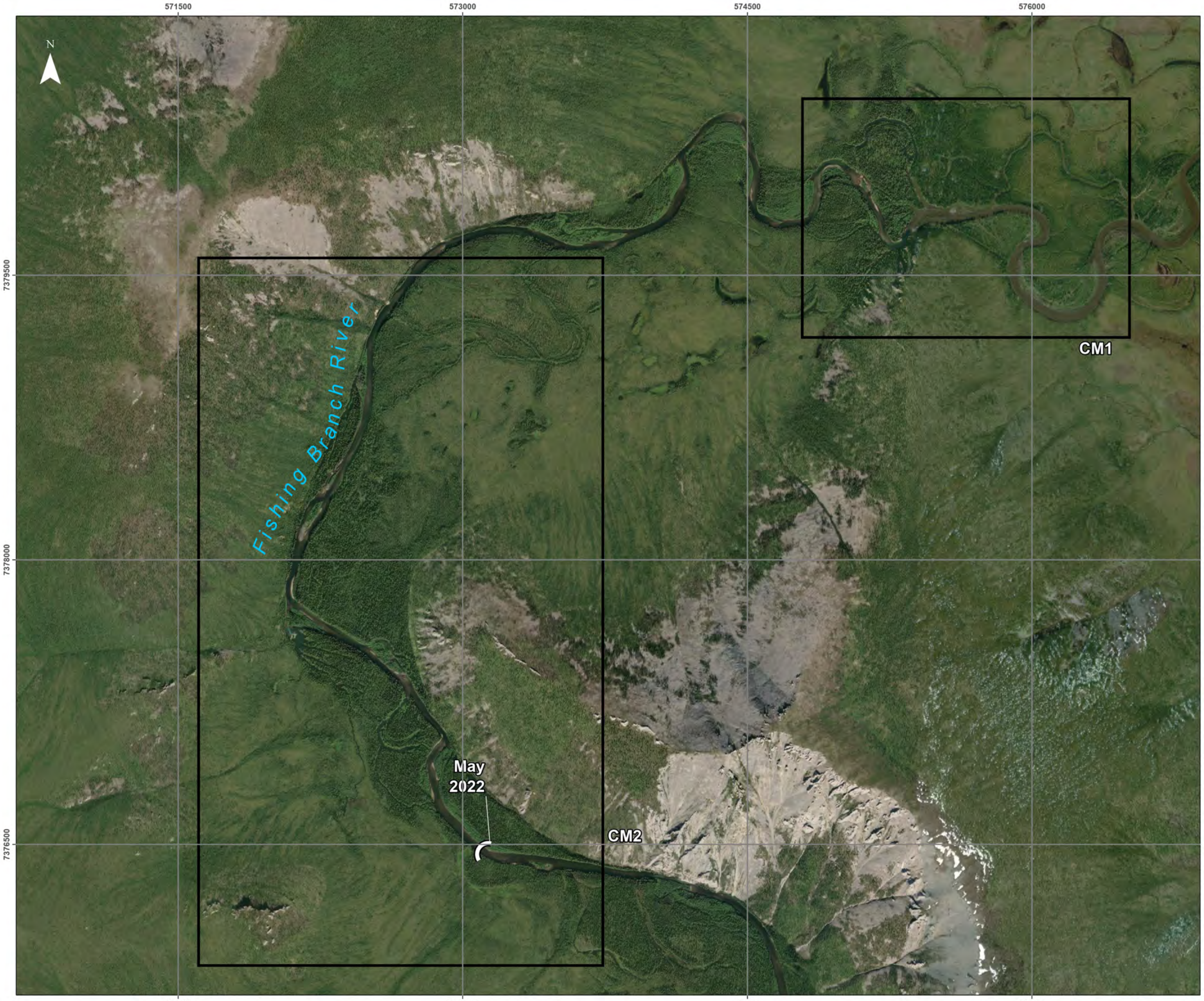
Drawn: PH	Checked: PSz	Map 5 - 1	Date: 2025-01-20

Vuntut Gwitchin First Nation

EDI

Path: L:\PROJECTS\2023\WH\23Y0250_VGFN_FishingBranch\23Y0250_mapping.aprx\23Y0250_Map5_DewateredExtents_series_20250120

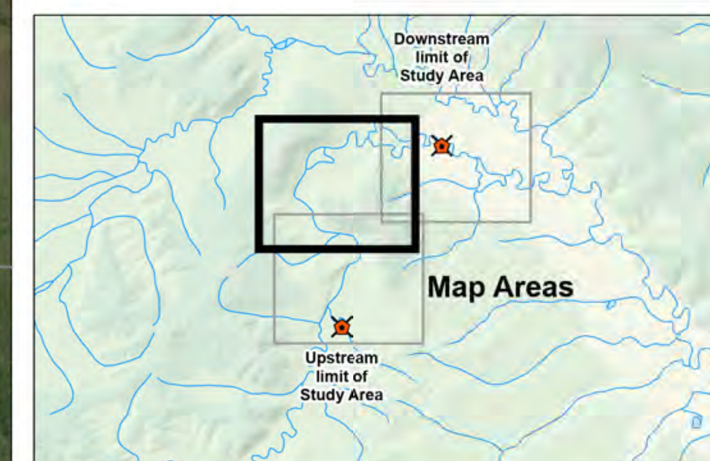


Fishing Branch River camera monitoring sites and spring dewatering locations

Map 2 of 3

Legend

- Limit of Study Area
- DFO Fishing Branch Weir
- Dewatered Extents
- Camera Monitoring Location



0 250 500 750 1,000
Metres

Map Scale = 1:23,000 (printed on 11 x 17)
Map Projection: NAD 1983 CSRS UTM Zone 7N

Data Sources
• Main Basemap. World Imagery: Earthstar Geographics
• Main data. CanVec 1:250,000 layers. Natural Resources Canada, 2024.
• Salmon observations. EDI Environmental Dynamics Inc, 2024.

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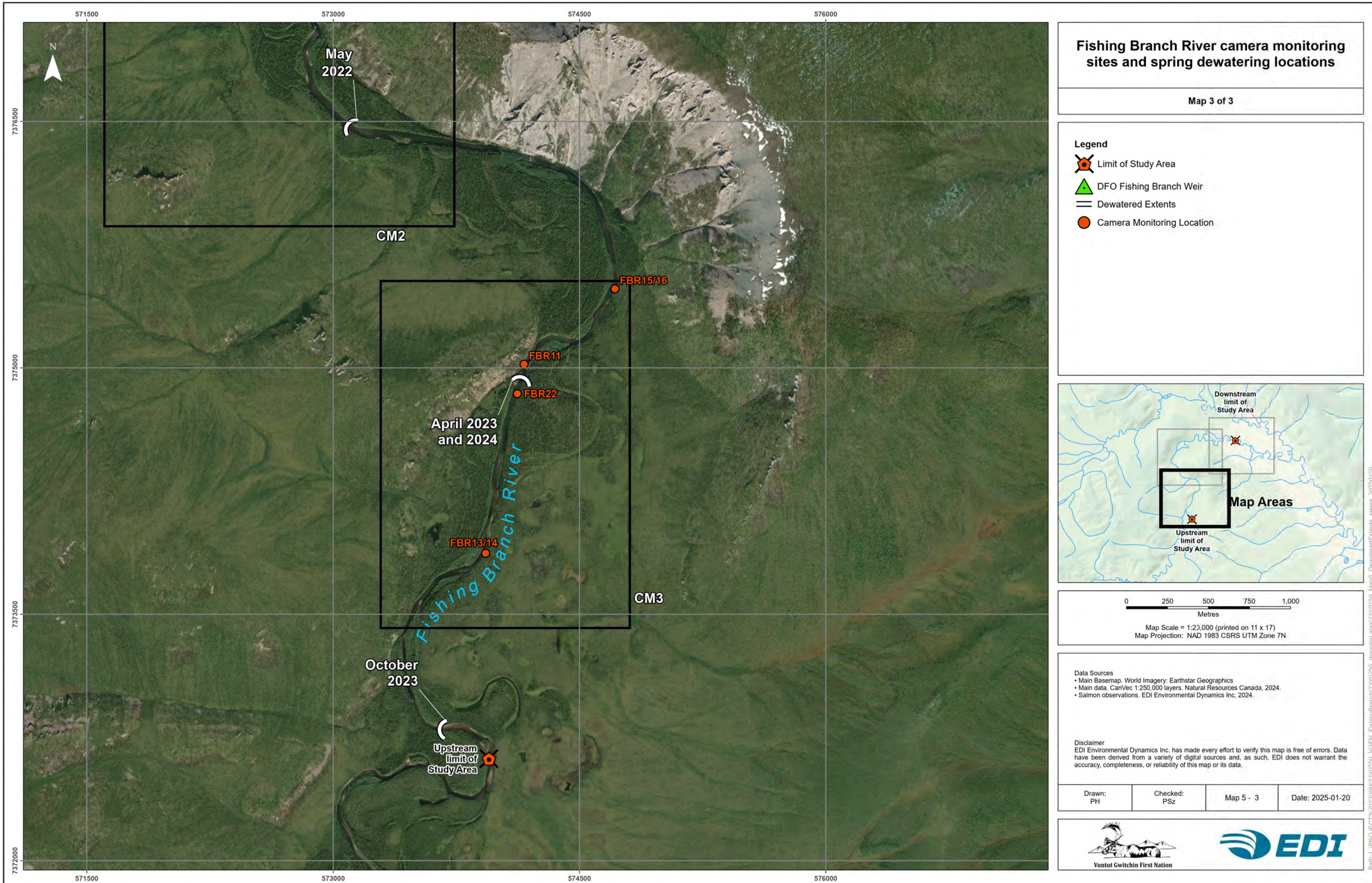
Drawn: PH	Checked: PSz	Map 5 - 2	Date: 2025-01-20

Vuntut Gwitchin First Nation

EDI


Fishing Branch River camera monitoring sites and spring dewatering locations
Map 3 of 3
Legend
Limit of Study Area
DFO Fishing Branch Weir
Dewatered Extents
Camera Monitoring Location
N
May 2022
CM2
CM3
FBR15/16
FBR11
FBR22
April 2023 and 2024
Fishing Branch River
FBR13/14
October 2023
Upstream limit of Study Area
571500
573000
574500
576000
7376500
7375000
7373500
7372000
Downstream limit of Study Area
Map Areas
Upstream limit of Study Area
0
250
500
750
1,000
Metres
Map Scale = 1:20,000 (printed on 11 x 17)
Map Projection: NAD 1983 CSRS UTM Zone 7N
Data Sources
• Main Basemap. World Imagery: Earthstar Geographics
• Main data. CanVec 1:250,000 layers. Natural Resources Canada, 2024.
• Salmon observations. EDI Environmental Dynamics Inc, 2024.
Disclaimer
EDI Environmental Dynamics Inc. has made every effort to verify this map is free of errors. Data have been derived from a variety of digital sources and, as such, EDI does not warrant the accuracy, completeness, or reliability of this map or its data.
Drawn: PH
Checked: PSz
Map 5 - 3
Date: 2025-01-20
Vuntut Gwitchin First Nation
EDI

3.2.1 MOST UPSTREAM CAMERAS – FBR13 AND FBR14

Cameras FBR13 (facing upstream) and FBR14 (facing downstream) are the farthest upstream remote cameras (Map 5), approximately 1.4 km downstream of where the river is already dewatered during spawning season (late October).

Photos of FBR13 are reported on in detail, as FBR14 was frequently moved by passing wildlife beginning in June 2022, and it was not accessed until October 28, 2023. The downloaded data shows that it was frequently shifted by wildlife between June 2022 until October 11, 2023, at which point wildlife destroyed the camera. It has since been mounted elsewhere, with plans to install lock boxes on all cameras in spring 2025.

3.2.1.1 Winter/spring dewatering in 2022 and 2023

During 2022 early spring, it was observed that this area became dewatered over the late winter months (Photo 6, Photo 7), reaching its maximum extent on May 11, 2022. Returning water was first observed at cameras FBR13 and FBR14 on May 12, 2022, flooding the banks rapidly within 24 hr (Photo 8 to Photo 10; Appendix Photo 1 to Appendix Photo 4).

Photo 6. Maximum dewatered extent at camera FBR13 (facing upstream) on May 11, 2022.

Photo 7. Maximum dewatered extent at camera FBR14 (facing downstream) on May 11, 2022.

Photo 8. First sign of flowing water returning at camera FBR13, May 12, 2022.

Photo 9. First sign of flowing water returning at camera FBR14, May 12, 2022.

Photo 10. Flowing water returned at camera FBR13 on May 13, 2022.

Photo 11. Fully wetted banks at camera FBR13 on May 14, 2022.

Flow continued in this area as expected throughout the spring, summer, and fall. Approximately 1.4 km upstream of these cameras, the river was dewatered by the end of October 2022, and this dewatered extent extended further downstream as winter progressed. By the beginning of March 2023, the flow in this area was greatly reduced (Photo 12), completely dewatering and eventually freezing over on March 24, 2023 (Photo 13).

This area of the river remained dewatered into the early spring, with the maximum dewatered extent occurring on May 10, 2023 (Photo 14), and water returning by May 11, 2023 (Photo 15). The return of water to Fishing Branch is very dynamic, with large changes to the flow of the area within 24 hr of the first signs of water captured on camera (Photo 16; Appendix C). The return of water in 2023 was one day before the return in 2022.

Photo 12. Reduced flows visible at camera FBR13 on March 1, 2023.

Photo 13. No flow visible at camera FBR13 on March 24, 2023.

Photo 14. **Maximum dewatered extent at camera FBR13 on May 10, 2023.**

Photo 15. **First sign of flowing water returning at camera FBR13, May 11, 2023.**

Photo 16. Flood conditions and surge of ice moving downriver at camera FBR13 on May 12, 2023, approximately 26 hr after the first signs of returning flow were recorded.

3.2.1.2 Winter/spring dewatering in 2024

Similar to 2022 and 2023, the wetted portion of this location was still relatively open as of February 24, 2024 (Photo 17), and rapidly dewatered by March 16, 2024 (Photo 18).

Photo 17. Open water at camera FBR13 on February 24, 2024.

Photo 18. Riverbed fully dewatered at camera FBR13 on March 16, 2024.

Of critical note for interpreting these results, in October 2023 there was a total of 267 redds enumerated upstream of FBR13 (i.e., in the 1.4 km between FBR13 and upstream where the river was dewatered in October 2023). Camera FBR14 was installed facing an area that had 21 redds in October 2023. As such, these cameras captured the dewatering of approximately 288 redds over the course of the late winter/early spring of 2024 (QAQC'd value = 187 redds).

This area remained dewatered throughout the late winter and early spring, reaching its maximum dewatered extent by May 14, 2024 (Photo 19). The return of water in spring 2024 was rapid, with flood conditions occurring less than 24 hr later, on May 15, 2024 (Photo 20). The following day, May 16, 2024, there was a large movement of ice and debris for a brief duration that the cameras fortunately captured (Photo 21); conditions returned to being relatively ice-free the following hour (Photo 22).

The changes to flow in Fishing Branch River throughout the year are very dynamic and interesting; see Appendix C for a more detailed time series of FBR13 between October 2023 and October 2024. Dates and times are provided in the top left of each image.

Photo 19. Maximum dewatered extent at camera FBR13 on May 14, 2024; 267 redds were enumerated upstream of FBR13 during the October 2023 aerial surveys.

Photo 20. Flood conditions at camera FBR13 on May 15, 2024.

Photo 21. A surge of ice and debris 25 hr after the first signs of water at camera FBR13, on May 16, 2024.

Photo 22. No ice or debris present at FBR13, one hour after high debris flows, on May 16, 2024.

Only photos of FBR13 are reported on in 2023 and 2024, as FBR14 was pointing in an unhelpful direction due to either wildlife or time constraints on reinstallation.

3.2.2 CENTRAL CAMERAS – FBR AND FBR22

Cameras FBR22 (facing upstream) and FBR11 (facing downstream) are located between the other two camera locations (approximately where the dewatered extent reaches most years; Map 5), 1.7 km upstream of Bear Cave Mountain. FBR22 is reported on in spring 2022 and 2023, whereas FBR11 only has data from spring 2024 dewatering.

3.2.2.1 Winter/spring dewatering in 2022 and 2023

Camera FBR22 reached its full dewatered extent on May 12, 2022 (Photo 23), and flowing water was first observed on May 13, 2022 (Photo 24). Though the resolution is coarse, this location is approximately one day delayed from upstream FBR13/14, which had the first flowing water on May 12, 2022 (above: Photo 8, Photo 9).

Photo 23. Maximum dewatered extent at camera FBR22 on May 12, 2022.

Photo 24. First sign of flowing water at camera FBR22, May 13, 2022.

Flow continued in this area as expected throughout the spring, summer, and fall. Unfortunately, the camera was disturbed by wildlife, thus the overview of the river was lost, with photos instead focusing on the near shore (Appendix Photo 5, Appendix Photo 6).

This area of the river remained dewatered into the early spring, with the maximum dewatered extent occurring on May 10, 2023 (Appendix Photo 5), and water returning by May 11, 2023 (Appendix Photo 6). The return of water to Fishing Branch River is very dynamic, with large changes to the flow of the area within 24 hr of the first signs of water captured on camera (Appendix Photo 7). The return of water in 2023 was one day before the return in 2022. The timing of the return of flows to camera FBR22 in spring 2023 were the same as FBR13/14, located upstream (Section 3.2.1 above).

3.2.2.2 Winter/spring dewatering in 2024

Camera FBR22 was disturbed and pointing at an unknown location against the shore for the 2024 winter/spring dewatering, thus FBR11 will be reported on only (FBR22 was reinstalled on April 26, 2024).

FBR11 is located directly downstream of a groundwater fed spring (FBR22 is above this spring), thus in most years (except for spring 2022), there is a section along the left downstream facing bank that remains wetted over the late winter, though the other bank dewaters fully (Photo 25). Similar to the upstream cameras, this location was in flood conditions on May 15, 2024 (Photo 26).

Photo 25. Camera FBR11 on May 14, 2024, when all the upstream cameras were at their maximum dewatered extent.

Photo 26. Camera FBR11 in flood conditions, less than 24 hr later.

3.2.3 MOST DOWNSTREAM CAMERAS – FBR15 AND FBR16

Cameras FBR15 (facing upstream) and FBR16 (facing downstream) are the farthest downstream of the current remote cameras (Map 5), just upstream of the bend in the river at Bear Cave Mountain. FBR16 is facing Bear Cave Mountain. FBR15 had an error occur on data transfer, thus only spring 2024 is reported for FBR15. FBR16 is reported on between May 9, 2022 to May 16, 2024.

3.2.3.1 Winter/spring dewatering in 2022 and 2023

FBR16 is the lowest downstream camera of the currently set up cameras; the camera was first installed in April 2022, thus no comparisons to past years in this stretch is possible. However, coarse scale satellite imagery suggests that in past years, it remains at least partially wetted (imagery only goes back to 2017; European Space Agency 2025). The outcomes of the dewatering in spring 2022 are discussed in detail in the previous report submitted to the Yukon Rive Panel (CRE-163-30; EDI 2021). Spring 2022 was characterized by the largest dewatering event since the beginning of VGFN and EDI work on Fishing Branch in 2013, extending over 33 km upstream of Bear Cave Mountain. This assessment was supported by field visits/installed remote cameras, as well as by satellite imagery dated back to 2017 (European Space Agency 2025). The river was dewatered up to Bear Cave Mountain, with thousands of fish found stranded and perished (EDI 2021). At the time of the 2021 report, these cameras were not downloaded, therefore this section will report on the results of the 2022 dewatering timings as well as the 2023 spring conditions.

The 2021 report details the fish stranding and signs of predation on the isolated pools and dried river bed; however, FBR16 also captured scavenging by a grizzly bear (*Ursus arctos horribilis*) across several days; first seen on camera on May 9, 2022, and in most frames until May 12, 2022 (Photo 27, Photo 28). This was either the same bear revisiting over multiple days, or multiple bears at different times. These cameras only take six photos a day at set times, so it is very likely more scavenging occurred than what was captured.

Photo 27. Image of a scavenging grizzly bear at FBR16 (facing downstream) on May 9, 2022.

The last day of the maximum dewatered extent at FBR16 was May 12, 2022 (Photo 28). At the most upstream cameras (FBR13/14), water began returning on May 12, 2022 and the river was fully wetted by May 13, 2022 (above, Section 3.2.1). In the case of FBR16, the riverbed went from completely dry on May 12, 2022 (Photo 28) to fully wetted (Photo 29) in fewer than 24 hours, on May 13, 2022. Similar to FBR22 (Section 3.2.2), water returned suddenly at FBR15/FBR16, whereas the upstream cameras FBR13/14 had a day where water appeared to be slowly returning, before fully wetting the riverbed the following day (Section 3.2.1).

The dewatering of spring 2022 is the most advanced that has ever been recorded as part of this project (EDI 2021), and it is interesting to notice that the dynamics of the returning water differ at different locations. The camera sites are all approximately 1 km from each other. Cameras FBR13/14 (most upstream) are 1 km upstream of FBR11/22, and 2 km upstream of cameras FBR15/16 (most downstream; Map 5). Images from FBR13/14 determined that water returned to this section more slowly (i.e., >24 hr), whereas FBR11/22 and FBR 15/16 went from dry river bed to flood conditions within 24 hr.

Photo 28. **Maximum dewatered extent at camera FBR16 on May 12, 2022. A revisiting scavenging grizzly bear is in the right of the photo.**

Photo 29. **First sign of flowing water returning on camera FBR16, May 13, 2022.**

Flow continues in this area as expected throughout the spring, summer, and fall. Unlike upstream camera locations, this area remained wetted throughout the year, though flow varied, and was reduced by end of March (Photo 30). At this same time, the upstream cameras (specifically FBR13/14, and FBR22 which had data recorded for winter and spring 2023) were fully dewatered (above, sections 3.2.1 and 3.2.2).

Photo 30. Flow at FBR16 on March 25, 2023; at the same time, there was no flow at the most upstream cameras.

This flow continued to diminish throughout the late winter and early spring, though it never had a complete loss of flow. From visual assessment, it was at its lowest flow on May 10, 2023 (Photo 31) and had flow return on May 11, 2023 (Photo 32), the same day the upper camera sites had water returning. The river reached flood conditions by May 12, and had a large movement of ice and debris on May 13, 2023 (Photo 33). The same could be seen at the same time on the upstream camera FBR13 (Appendix Photo 8).

Photo 31. **Lowest flow observed at FBR16, occurring on May 10, 2023.**

Photo 32. **Increase in flow at FBR16 on May 11, 2023.**

Photo 33. Flood conditions and surge of ice moving downriver at camera FBR16 on May 12, 2023, 26 hr after the first signs of returning flow.

3.2.3.2 Winter/spring dewatering in 2024

Both FBR15 and FBR16 were non-operational between October 28, 2023 to April 27, 2024; as such, the winter conditions were not recorded. However, the April 26, 2024 field visit had them operational and able to capture the return of high flows in early May 2024. Camera FBR15 is reported on for 2024, as it had a better perspective; FBR14 had been slightly moved and did not capture as clear of an overview as it did in 2022 and 2023.

Similar to 2023, this area upstream of Bear Cave Mountain had continuous flow throughout the year, though much reduced, with lowest flows occurring on May 14, 2024 (Photo 34). High flows returned within 24 hr (Photo 35), and another 24 hr after that a large but brief pulse of ice and debris was captured (Photo 36). This pulse also occurred in the upstream cameras, and likewise, was brief and flushed through within 30 min (Photo 37).

Photo 34. Lowest flow observed at FBR15, occurring on May 14, 2024.

Photo 35. Camera FBR15 in flood conditions, less than 24 hr later, on May 15, 2024.

Photo 36. **Movement of ice and debris 25 hr after the return of high flows at FBR15, on May 16, 2024. The same ice and debris were observed in the upstream cameras.**

Photo 37. **Most ice and debris flushed out, 30 min following the large pulse on May 16, 2024.**

It is challenging to determine exactly which redds from October 2023 were affected by the reduced flows at FBR15 and FBR16; there were 16 located adjacent to the camera sites.

In future years, drone footage will be recorded from Bear Cave Mountain to the October dewatered extent 4.5 km upstream. This will be completed during the spawning season (October visit) and again during the dewatered visit (May). Using this footage, redd location and fine scale dewatering will be assessed to provide an estimate of redd loss in areas that don't fully dewater, but do experience reduced flows and areas of dewatered riverbed. Historically, the number of dewatered redds have been estimated only in the areas that experience complete dewatering. However, observing the exposed riverbed in early May at the sites with reduced flow (e.g., FBR15/16, FBR 11/22), it is likely that even if there are parts of the river that are wetted, other parts are exposed and redds are dewatered (e.g., Photo 25, Photo 28, Photo 31).

Species observed on the cameras are moose (*Alces alces*; Appendix Photo 9), grizzly bear (Photo 27), unidentified duck species (Appendix Photo 10) and unidentified songbird species, likely Canada jay (*Perisoreus canadensis*; Appendix Photo 11).

4 SUMMARY AND CONCLUSIONS

This report outlined the results from the October 2023 chum salmon spawning surveys, as well as remote camera data for 2022, 2023, and spring 2024. The purpose of the chum salmon spawning surveys is to get an estimate of the number of redds above the DFO weir, and then using remote camera data and a spring investigation, estimating the number of these redds that become fully dewatered over the late winter and early spring.

The results from the October 2023 aerial survey indicated a similar pattern that has been supported in past surveys, with three main clusters of chum redds above the DFO weir, referred to as CM1, CM2, and CM3 (downstream to upstream). Of these clusters, CM1 has never experienced dewatering as it is downstream of an aquifer adjacent to Bear Cave Mountain, ensuring flow throughout the year (Utting et al. 2012, EDI 2016). CM2 also typically does not experience dewatering, except for an anomalous dewatering event in May 2022, which extended in a small portion of CM2. However, at least a portion of the spawning cluster CM3 is at risk of dewatering every year (in 2022 all of CM3 was dewatered). In 2023, between 187 and 288 redds[1] were dewatered in CM3 (i.e., any redds upstream of the April 2024 dewatered extent), accounting for 12% of the redds enumerated during the October 2023 spawning survey (Map 3). In addition to the redds that are in areas that entirely dewatered, there are likely more redds between the dewatered extent and Bear Cave Mountain that become dewatered, as this stretch of river has significantly reduced flows over the winter and early spring than when chum spawn in the preceding fall. By October, the Fishing Branch River is already dewatered to the upper extent of the aerial surveys, approximately 4.5 km upstream of Bear Cave Mountain (Photo 5).

The remote camera data was intended to provide information on the advance of the dewatered extent throughout the winter and early spring. This report provided data on the 2022, 2023, and 2024 spring dewatered extent, as well as winter conditions of reduced flows. The highest upstream cameras (FBR13/14) were dewatered in every year of remote camera data, as they are farthest upstream (and closest to the October dewatered extent, which advances throughout the winter; Map 5). The centrally located cameras, FBR11/22 are located upstream of a groundwater spring (FBR22), and downstream of the same the groundwater spring (FBR11; Map 5). FBR11 and FBR22 were both fully dewatered in 2022, which is considered an anomalous year of dewatering as it exceeded anything in project history. In 2023 and 2024, FBR22 was fully dewatered, whereas FBR11 still had flow along the left downstream bank due to its proximity to the groundwater spring. The farthest downstream cameras, FBR15 and FBR16, were also both completely dewatered in the spring of 2022 (detailed more in EDI 2021). In 2023 and 2024, the flows were greatly reduced throughout the winter and early spring, exposing some of the riverbed, though some of the river remained wetted.

Most years of remote camera data were characterized by a sudden return of flows, usually less than 24 hr from being fully dewatered to being completely wetted and flowing. The river typically reached flood conditions within 24 hr of the first signs of flow. In the years it was captured, another 24 hr after water returned, the river was characterized by high flows and a brief pulse of large amounts of woody debris and ice/snow, which

[1] 187 redds represent the QAQC'd value, and 288 represent the on the ground value; regardless of which is used, they are proportions thus both make up 12% of the redds enumerated.

flushed out within 30 min to 1 hr. The predictability of how fast water returns and this repeated annual surge of ice and debris 48 hr after the lowest flows is an interesting finding and one that deserves further investigation.

The dynamic processes which have the Fishing Branch River go from a dry riverbed to flood conditions within 24 hrs are thus far, poorly understood. Due to the extreme dewatering event that was captured in the spring of 2022, the results of this project were presented to a diverse audience of experts from governmental and academic agencies in 2023. An outcome of this presentation was the formation of a panel to provide their expertise to better study and understand the dynamics responsible for the annual dewatering occurrences on Fishing Branch River. This group consists of VGFN members, EDI employees, and faculty members from Queen's and Brock universities, as well as members of the Yukon Water Resources Branch. Yukon Parks and DFO are also logistical collaborators on the project. It is the intention of this group to expand the project in 2024/2025 to include more water sampling, hydrology and monitoring. In addition to a pending application with the Yukon River Panel, this group has successfully secured funding from the Climate Change Preparedness in the North Program to partially fund future initiatives regarding components of the project focused on hydrogeology and river dynamics of this system. This project will also continue as it is in its current form, focusing on spawning chum and the dewatered extents, and estimating the number of redds dewatered annually.

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APPENDICES

APPENDIX A CHUM SALMON REDDS, SPAWNERS, AND CARCASSES OBSERVED DURING OCTOBER 2023 AERIAL SURVEY

Appendix Table 1. Results for the chum salmon aerial survey on October 28, 2023.

Latitude	Longitude	Number of Redds	Number of Spawners	Number of Carcasses	Comments
66.527521	-139.246456	-	-	-	Start of survey
66.527521	-139.246456	0	1	2	
66.528721	-139.247500	0	0	6	
66.529929	-139.249201	1	0	0	
66.531483	-139.251571	0	0	5	
66.530684	-139.257686	1	3	3	
66.529202	-139.255959	0	0	1	
66.527526	-139.258938	0	0	28	
66.527274	-139.260583	0	3	12	
66.527544	-139.262212	0	0	3	
66.528448	-139.263433	8	0	1	
66.529208	-139.262128	0	0	19	
66.530032	-139.260204	0	0	17	
66.531695	-139.263842	0	0	69	
66.531263	-139.268068	0	0	20	
66.530232	-139.271258	0	0	17	
66.529312	-139.272472	0	0	5	
66.527952	-139.270574	0	1	32	
66.526780	-139.271411	0	0	20	
66.526630	-139.272228	7	1	19	
66.527206	-139.276367	5	0	6	
66.527746	-139.278902	3	0	21	
66.526826	-139.282308	0	1	69	
66.524798	-139.281895	0	0	11	
66.523824	-139.283044	0	0	7	
66.523346	-139.284880	0	0	14	
66.523401	-139.287920	45	0	6	
66.523764	-139.288888	22	5	15	
66.524224	-139.290828	65	11	17	
66.524175	-139.291415	37	19	31	
66.525843	-139.292596	24	11	21	
66.526567	-139.291010	5	0	16	
66.527419	-139.288100	1	1	17	
66.528509	-139.293474	0	3	16	
66.528298	-139.299876	4	4	50	
66.527786	-139.300588	49	90	250	
66.528413	-139.301851	370	17	36	
66.530476	-139.303955	110	171	104	

Latitude	Longitude	Number of Redds	Number of Spawners	Number of Carcasses	Comments
66.527521	-139.246456	-	-	-	Start of survey
66.527527	-139.307324	346	207	101	
66.530983	-139.311715	170	64	6	
66.529047	-139.315374	11	0	2	2 wolves
66.528772	-139.321990	0	0	27	
66.531431	-139.331608	0	0	33	
66.527535	-139.344223	0	0	46	
66.528350	-139.352263	0	0	9	
66.528060	-139.356360	0	0	29	
66.525979	-139.363946	4	3	26	
66.523138	-139.368765	0	0	42	
66.520093	-139.369446	6	4	19	
66.518938	-139.370362	46	6	20	
66.515220	-139.375430	83	3	72	
66.512029	-139.379174	230	122	56	
66.508914	-139.372809	44	23	26	
66.507726	-139.369007	67	44	60	
66.505565	-139.364068	120	150	17	
66.504095	-139.361244	110	60	39	
66.500284	-139.361107	51	37	5	
66.498068	-139.349577	18	15	0	
66.496732	-139.332946	0	0	7	
66.492554	-139.322715	7	6	5	
66.490769	-139.322202	9	5	2	
66.486901	-139.327713	26	20	4	
66.485588	-139.332984	9	15	15	
66.480958	-139.338100	21	26	14	
66.478300	-139.339355	30	23	8	
66.474763	-139.342500	35	27	11	
66.473494	-139.347891	202	230	23	
66.465948	-139.347367	-	-	-	End of survey
Total		**2402**	**1432**	**1710**	

APPENDIX B REMOTE CAMERA PHOTOS

Appendix Photo 1. First sign of flowing water on camera FBR14, May 12, 2022 (12:00).

Appendix Photo 2. Water advance within 2 hr at camera FBR14, May 12, 2022 (14:00).

Appendix Photo 3. 24 hr after the first signs of water at camera FBR14, on May 13, 2022 (12:00).

Appendix Photo 4. 48 hr after the first signs of water at camera FBR14, on May 14, 2022 (12:00).

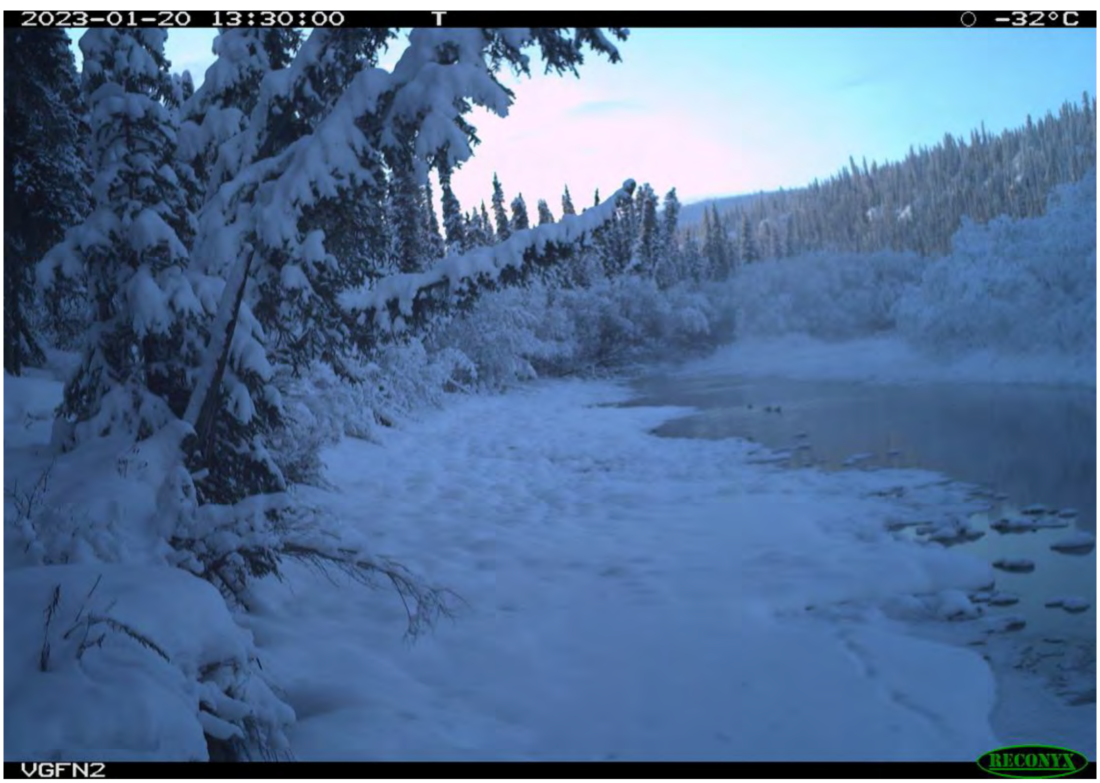
Appendix Photo 5. **FBR22 showing the maximum dewatered extent visible, on May 10, 2023.**

Appendix Photo 6. **First sign of flowing water on camera FBR22 May 11, 2023 (12:00).**

Appendix Photo 7. **24 hr after the first signs of water at camera FBR22, on May 12, 2023 (12:00).**

Appendix Photo 8. **Surge of ice and debris on camera FBR13, May 13, 2023.**

Appendix Photo 9. Moose captured at FBR13 of February 21, 2023.



Appendix Photo 10. Unidentified duck species captured at FBR13, January 20, 2023 in -32 °C.

Appendix Photo 11. Unidentified songbird species, likely Canada jay, captured at FBR13, March 26, 2023.

APPENDIX C CAMERA FBR13 FALL 2023 TO FALL 2024


2023-11-17 15:00:00 T
-12°C
VGFN2
RECONYX


2023-12-17 15:00:00 T
-25°C
VGFN2
RECONYX


2024-01-17 15:00:00 T
-22°C
VGFN2
RECONYX


2024-02-17 15:00:00 T
-20°C
VGFN2
RECONYX


2024-03-17 15:00:00 T
-19°C
VGFN2
RECONYX


2024-04-17 15:00:00 T
10°C
VGFN2
RECONYX


2024-05-14 15:00:00 T
12°C
VGFN2
RECONYX


2024-05-15 12:00:00 T
11°C
VGFN2
RECONYX


2024-05-15 13:00:00 T
12°C
VGFN2
RECONYX


2024-05-15 14:00:00 T
14°C
VGFN2
RECONYX


2024-05-15 15:00:00 T
10°C
VGFN2
RECONYX


2024-05-16 12:00:00 T
14°C
VGFN2
RECONYX


2024-05-16 13:00:00 T
16°C
VGFN2
RECONYX


2024-05-16 14:00:00 T 18°C
VGFN2
RECONYX


2024-05-16 15:00:00 T
14°C
VGFN2
RECONYX


2024-05-17 12:00:00 T 13°C
VGFN2
RECONYX


2024-05-17 13:00:00 T
14°C
VGFN2
RECONYX

2024-05-17 14:00:00 T
15°C
VGFN2
RECONYX


2024-05-17 15:00:00 T
15°C
VGFN2
RECONYX


2024-06-17 15:00:00 T
20°C
VGFN2
RECONYX


2024-07-17 15:00:00 T
14°C
VGFN2
RECONYX


2024-08-17 15:00:00 T
14°C
VGFN2
RECONYX


2024-09-17 15:00:00 T
7°C
VGFN2
RECONYX


2024-10-17 15:00:00 T
-5°C
VGFN2
RECONYX


2024-10-31 15:00:00 T
-14°C
VGFN2
RECONYX